Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 1

Date: 04-03-2026

Academic Campus Event & Registration Portal

SDG 4 – Quality Education | SDG 9 – Industry, Innovation and Infrastructure

Presented By:

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Courses Handling Faculty:

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INTRODUCTION

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- Academic institutions host numerous events – workshops, seminars, sports and cultural fests – but lack a unified digital system for discovery and registration.
- Manual registration processes cause data loss, overbooking, and poor communication between organizers and participants.
- This project presents a full-stack Academic Campus Event & Registration Portal built using Java Spring Boot, MySQL, and HTML/CSS/JavaScript.
- Students can browse upcoming events, register or cancel participation, and view their registration history in real time.
- Admins can create, update, and manage events with capacity controls and registration deadlines via REST APIs.
- The system supports role-based access for Students, Faculty, and Admins with Spring Security and BCrypt authentication.
- A dedicated Admin Dashboard allows admins to create new events, view all registrations per event, and manage listings from a single interface.
- Registered participants can download a personalized PDF Participation Certificate after successful event registration, with event name, date, and student details auto-filled.
- Aligns with SDG 4 (Quality Education) and SDG 9 (Innovation & Infrastructure) by promoting structured academic engagement.

PROBLEM STATEMENT

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- Campus event management is currently handled through manual processes, paper-based registrations, and informal channels such as notice boards and WhatsApp groups.
- There is no centralized platform where students can discover all campus events, check seat availability, and register online – leading to missed opportunities and overbooking.
- Admins lack tools to manage event capacity, monitor registrations in real time, or enforce registration deadlines, causing administrative overhead and data errors.
- Students have no way to track their own registration history or receive confirmation, leading to confusion and duplicate registrations.
- There is a critical need for a secure, role-based, web-based portal that automates event publishing, student registration, and data management for academic institutions.

OBJECTIVE(S)

- To design and develop a full-stack Academic Campus Event & Registration Portal that enables students and faculty to discover, register, and manage campus events through a centralized web application.
- To implement a secure, role-based backend using Java Spring Boot and MySQL that supports event CRUD operations, real-time seat tracking, registration deadline enforcement, and user authentication via Spring Security.
- To build a dedicated Admin Dashboard for streamlined event creation and management, and a Certificate Generation module that produces downloadable PDF participation certificates for registered students.

PROPOSED SYSTEM

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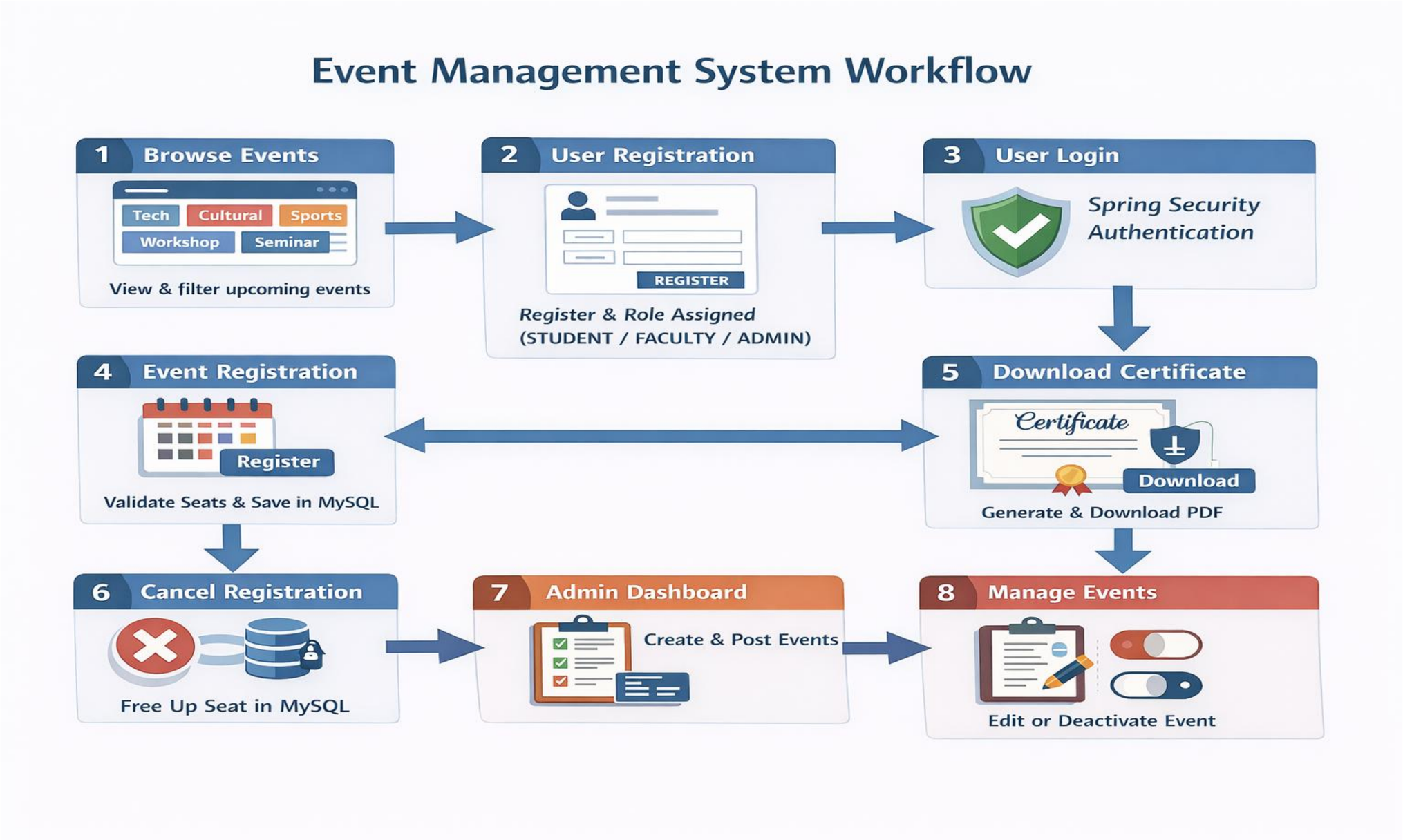
- A web-based Campus Event & Registration Portal where students can browse, filter by category (Tech / Cultural / Sports / Workshop / Seminar), and register for events.
- Role-based access control: Students register or cancel participation; Admins create, update, and delete events with full capacity and deadline management.
- Real-time seat availability tracking prevents overbooking; registration deadline enforcement automatically blocks late sign-ups.
- Spring Boot REST API backend with JPA/Hibernate maps to a MySQL relational database with four normalized tables: Users, Events, Registrations, and Certificates.
- Spring Security with BCrypt password encoding ensures secure login and eliminates the need for manual or paper-based registration.
- Admin Dashboard: A protected interface for creating new events (form-based), viewing all registrants per event, editing event details, and toggling event active/inactive status.
- Certificate Generation: On confirmed registration, students can request and instantly download a personalized PDF participation certificate auto-filled with their name, event title, date, and venue.

PROPOSED SYSTEM ARCHITECTURE

WORKFLOW


Event Management System Workflow
1 Browse Events
Tech
Cultural
Sports
Workshop
Seminar
View & filter upcoming events
2 User Registration
REGISTER
Register & Role Assigned
(STUDENT / FACULTY / ADMIN)
3 User Login
Spring Security
Authentication
4 Event Registration
Register
Validate Seats & Save in MySQL
5 Download Certificate
Certificate
Download
Generate & Download PDF
6 Cancel Registration
Free Up Seat in MySQL
7 Admin Dashboard
Create & Post Events
8 Manage Events
Edit or Deactivate Event

TECHNOLOGIES USED

- **Frontend:** HTML5, CSS3, Vanilla JavaScript (Fetch API, DOM manipulation, Local Storage for session state)
- **Backend:** Java 17, Spring Boot 3.2, Spring MVC, Spring Data JPA, Spring Security, Lombok, iText / Apache PDFBox (PDF Certificate Generation)
- **Database:** MySQL 8.0 (Relational DB with 4 normalized tables), Hibernate ORM, MySQL Workbench for DB administration
- **Tools:** Eclipse IDE (Enterprise Java), Apache Maven (build & dependency management), Git (version control), Postman (API testing)
- **Hardware / Software Requirements:** OS: Windows 10/11 or Linux | RAM: 8 GB minimum | JDK 17+ | MySQL Server 8.0+ | Browser: Chrome or Firefox

MODULE DESCRIPTION

Module 1: User Management

Handles student/faculty/admin registration, login, and profile management. Uses Spring Security with BCrypt for secure password storage and role-based access control (STUDENT, FACULTY, ADMIN).

Module 2: Event Management

Admin can create, update, and soft-delete events. Supports categories (Tech, Cultural, Sports, Workshop, Seminar), venue, date, max capacity, registration deadline, and organizer details via REST APIs.

Module 3: Registration & Seat Management

Students register or cancel event participation. System validates seat availability and deadline before confirming. Real-time seat count is updated transactionally in MySQL on every registration or cancellation.

Module 4: Database Management

MySQL with 4 normalized tables: users, events, registrations, certificates. JPA/Hibernate handles ORM mapping; Spring Data JPA repositories manage CRUD, search queries, and unique constraint enforcement.

Module 5: Admin Dashboard

A protected admin-only interface for creating new events (form-based UI), viewing all registrations per event, editing event details, and activating or deactivating events. Accessible only to ADMIN role users.

Module 6: Certificate Generation

On confirmed registration, students request and download a personalized PDF participation certificate. Backend auto-fills student name, event title, date, and venue using iText/Apache PDFBox and serves the file as a direct download.

IMPLEMENTATION SCREENSHOTS

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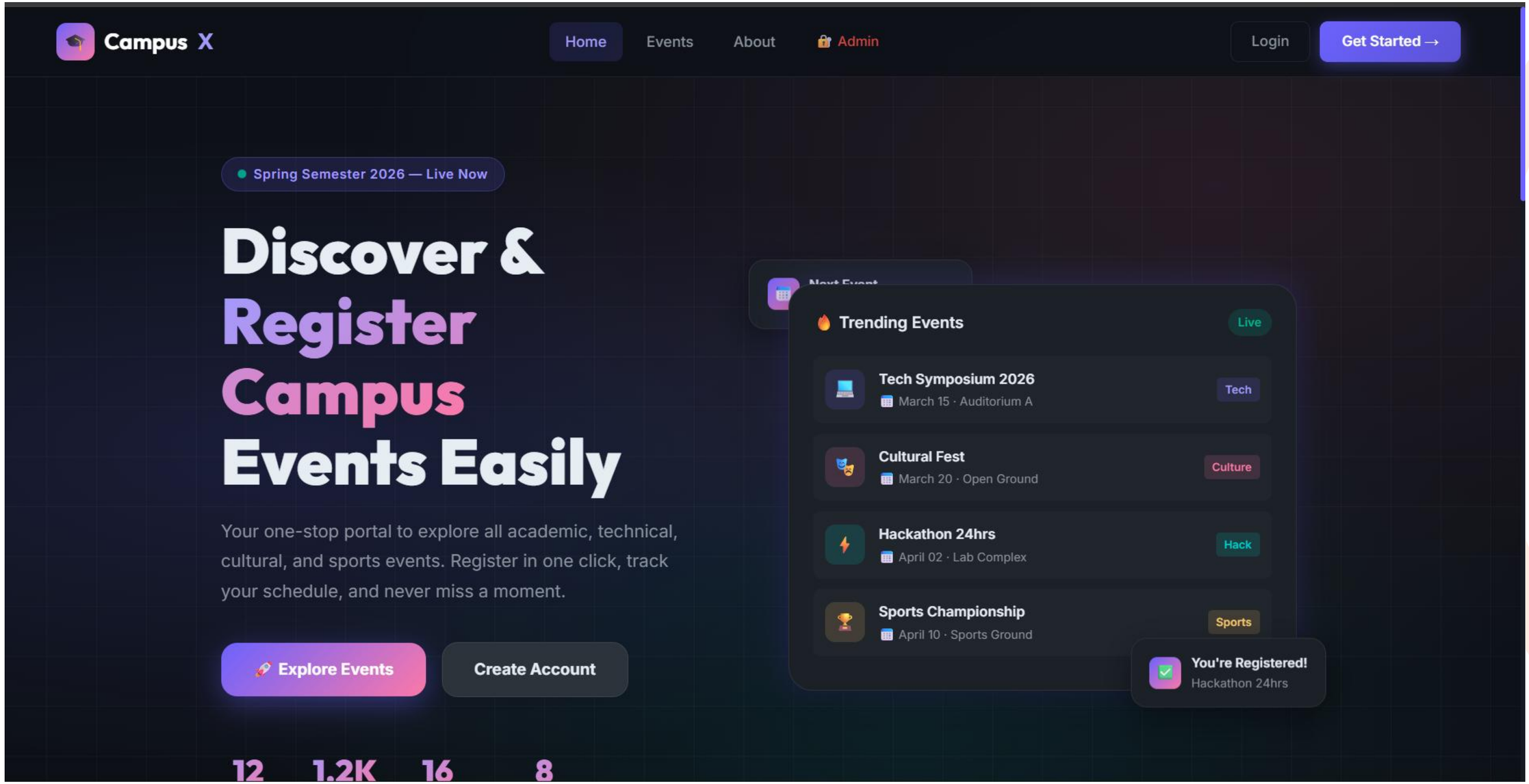

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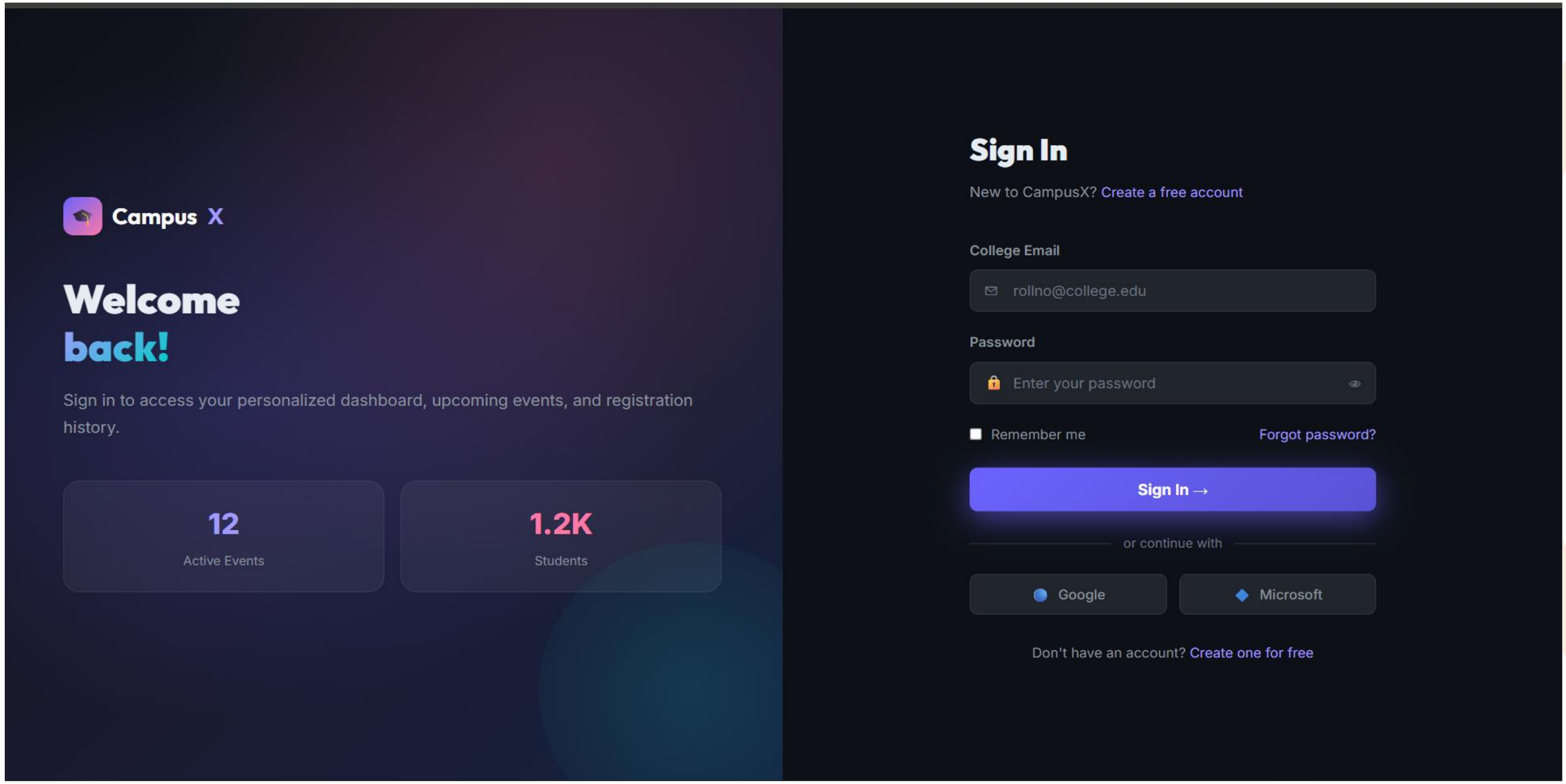

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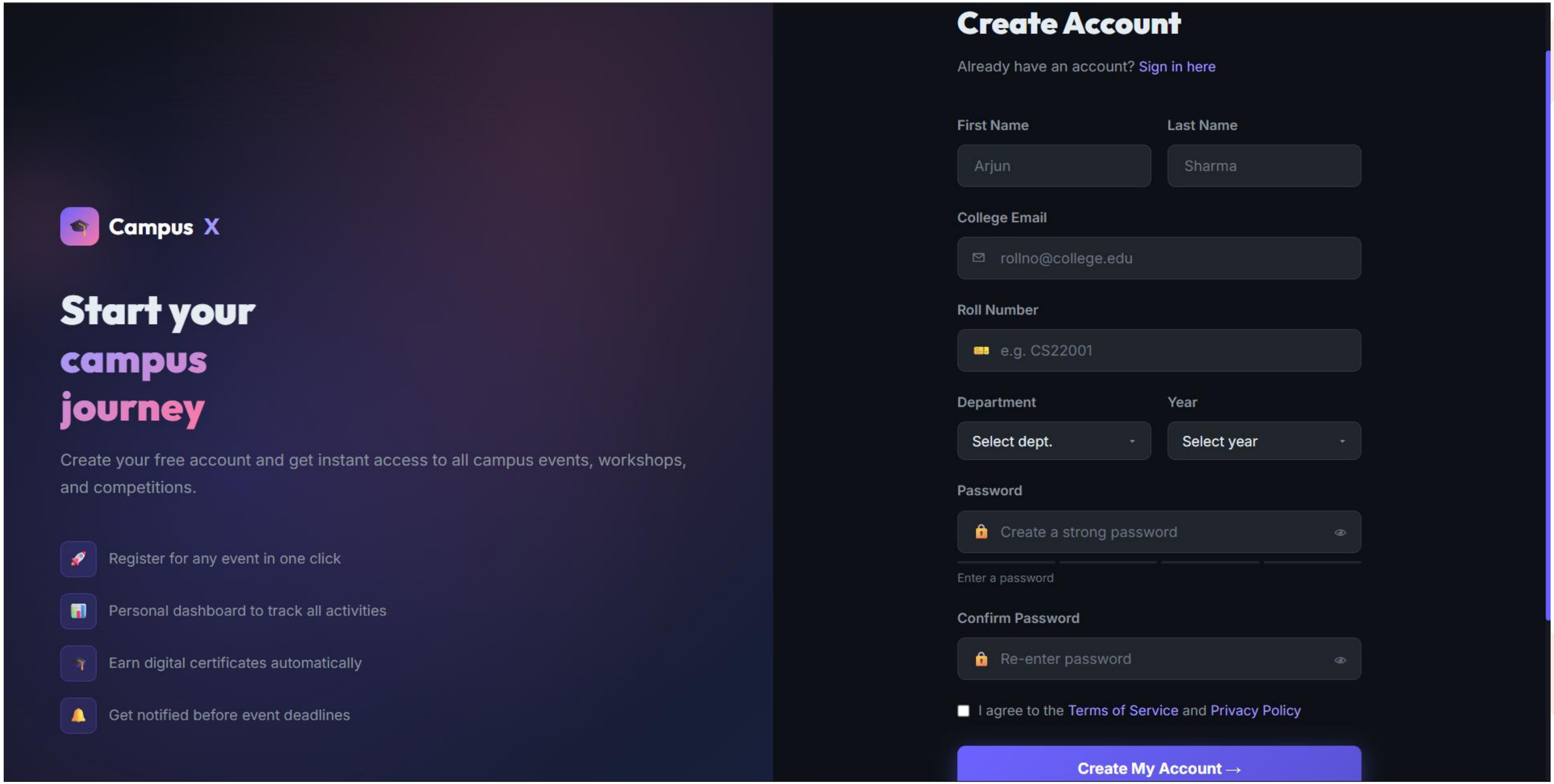

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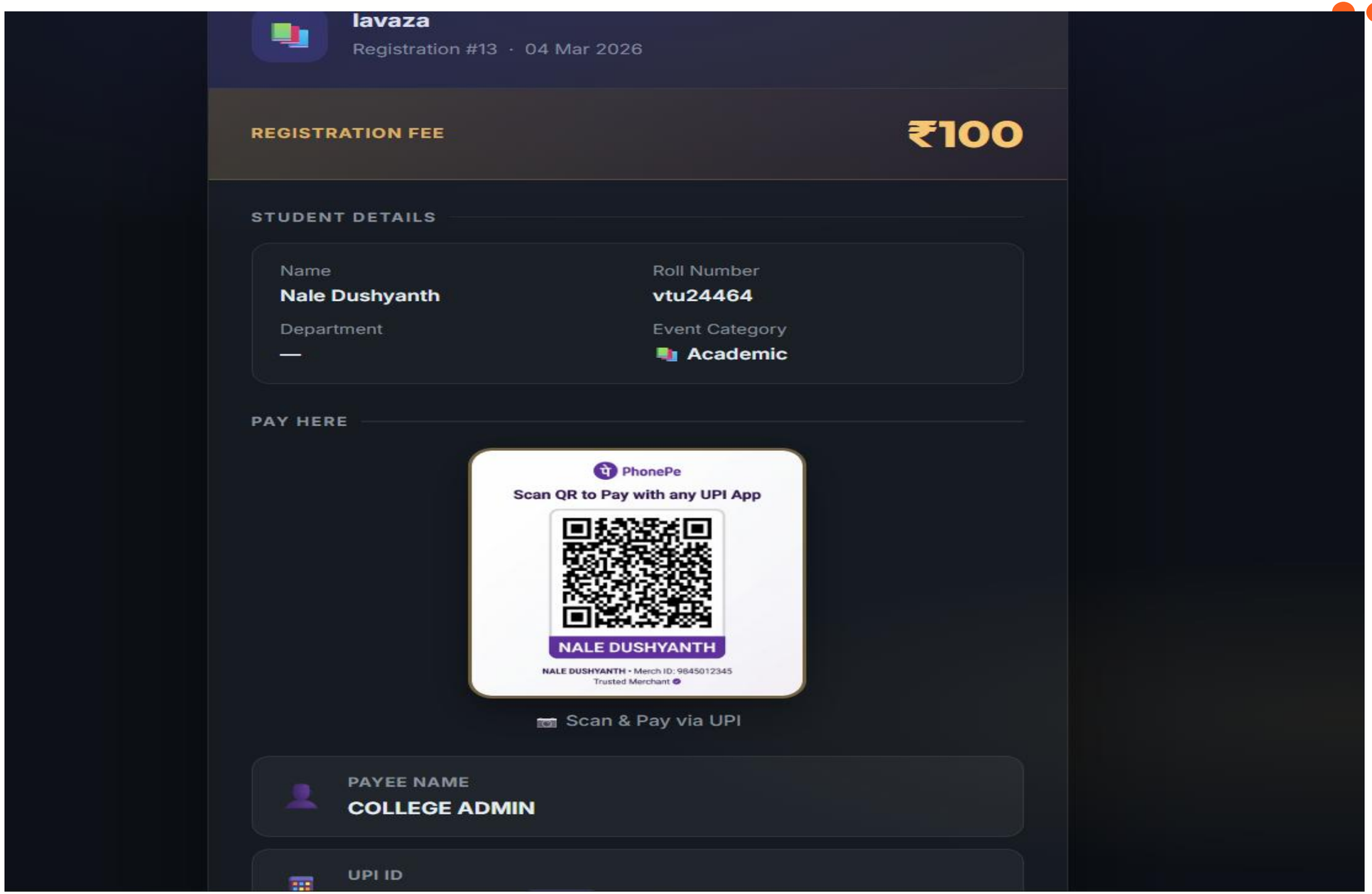

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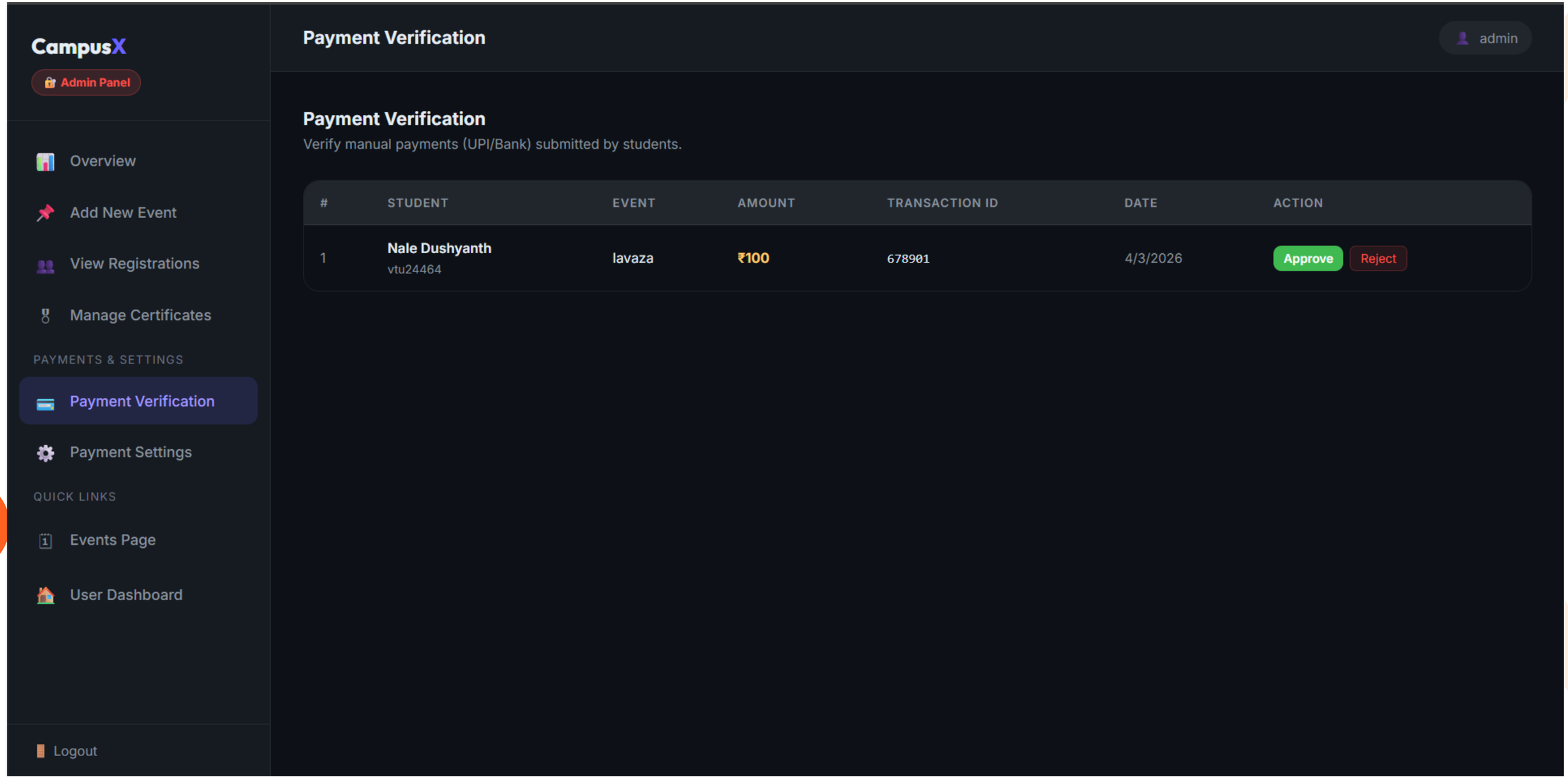

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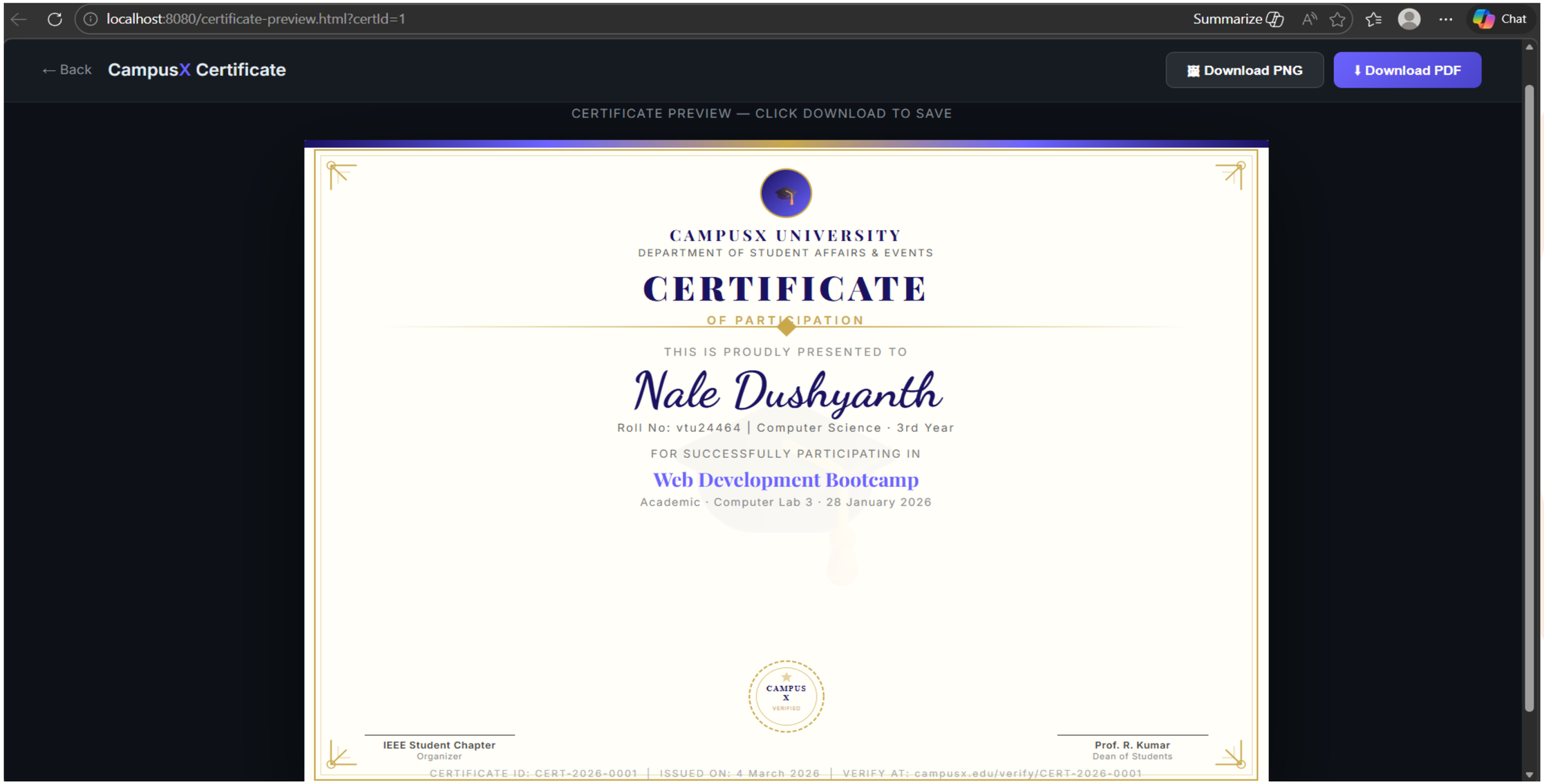

ADVANTAGES & LIMITATIONS

ADVANTAGES:

- Centralized platform eliminates manual/paper-based registration and significantly reduces administrative workload.
- Real-time seat tracking and deadline enforcement prevent overbooking and late registrations automatically.
- Role-based access (Student/Faculty/Admin) with Spring Security ensures data privacy and appropriate access control for all users.
- RESTful API design enables easy extension and future integration with mobile apps or external institutional systems.
- Admin Dashboard centralizes event creation and management, removing dependency on external tools or manual coordination.
- Downloadable PDF Certificate adds tangible value for students – providing official, auto-generated proof of campus event participation.

LIMITATIONS:

- Currently a web-only application; no dedicated Android/iOS mobile app is available at this stage.
- Email notification and reminder feature for registered participants has not yet been implemented.
- No QR code-based attendance tracking or check-in system is integrated currently; planned for a future milestone.

Thank you